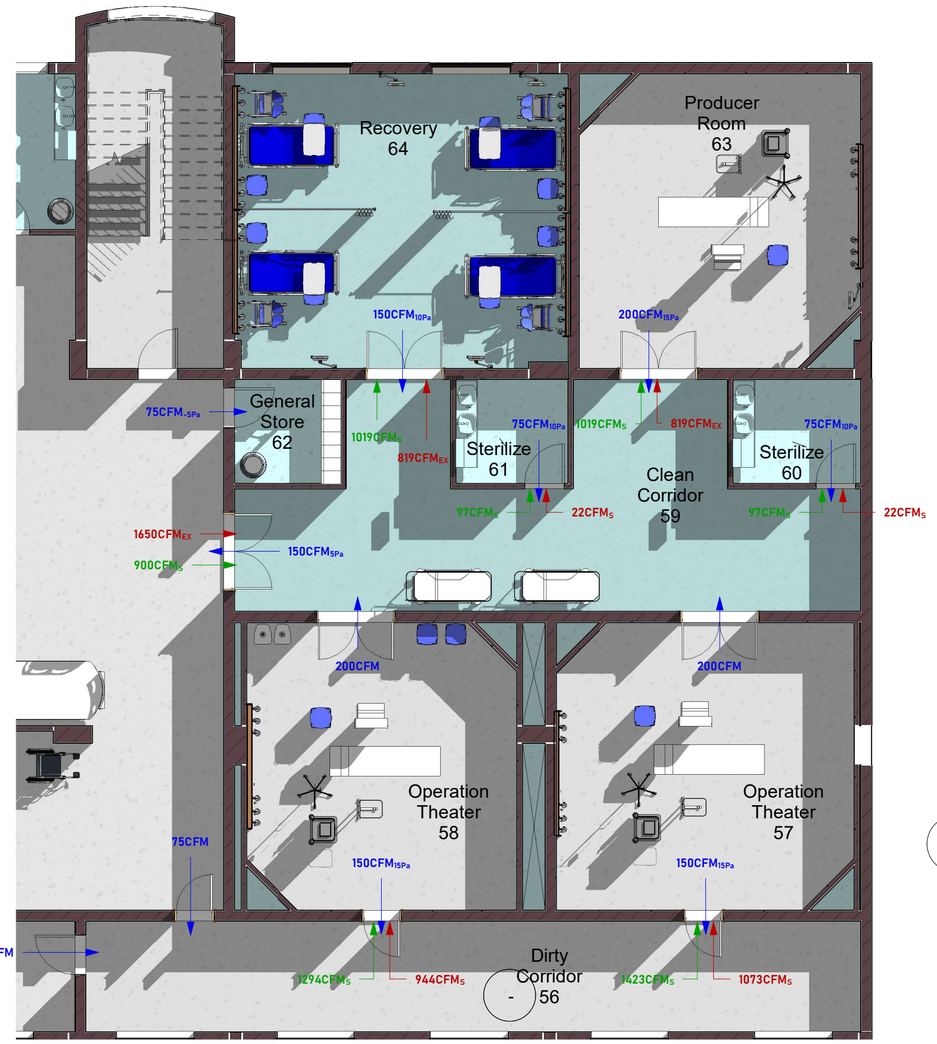
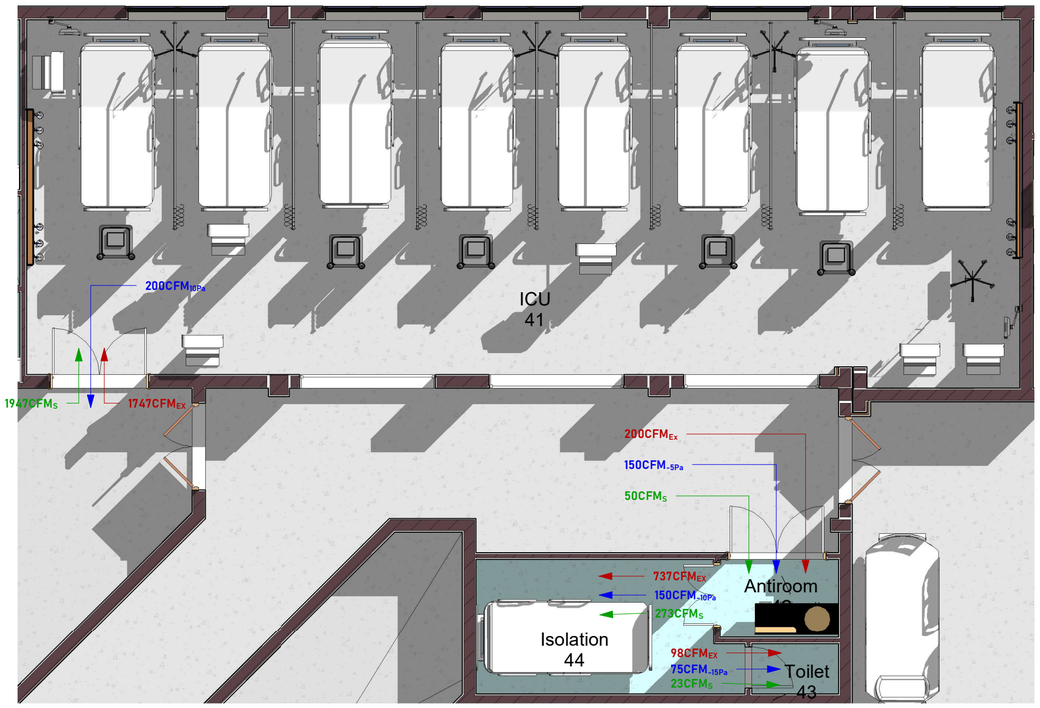


1 Ground Floor Critical Areas1

1 : 100

Space Name	Space Number	Area(m^2)	Height (m)	ACH	CFM_S	CFM_{EX}	Pressure(pa)
Producer Room	63	42	2.75	15	1019	819	(+15)
Recovery Room	64	52	2.75	8	672	522	(+10)
Operation Theater	57	44	2.75	20	1423	1073	(+15)
Operation Theater	58	40	2.75	20	1294	944	(+15)
Sterilize	60	7	2.75	10	113	38	(+10)
Sterilize	61	6	2.75	10	97	22	(+10)
Clean Corridor	59	56	2.75	10	900	1650	(+5)
Dirty Corridor	56	45.5	2.75	4	0	450	(-5)
General Store	62	5.9	2.75	10	0	75	(-5)



2 Ground Floor Critical Areas2

1 : 80

Space Name	Space Number	Area(m^2)	Height (m)	ACH	CFM_S	CFM_{EX}	Pressure(pa)
Antiroom	42	3.1	2.75	10	50	200	(-5)
solation	44	11.5	2.75	12	223	373	(-10)
ICU	41	120.4	2.75	10	1947	1747	(+10)
Toilet	43	1.48	2.75	10	23	98	(-15)

Table 7.1 Design Parameters—Inpatient Spaces—Hospital Spaces

Function of Space (ad)	Pressure Relationship to Adjacent Areas (n)	Minimum Outdoor ach	Minimum Total ach	All Room Air Exhausted Directly to Outdoors (j)	Air Recirculated by Means of Room Units (a)	Unoccupied Turndown	Minimum Filter Efficiencies (ab)	Design Relative Humidity (k), %	Design Temperature (l), °F/°C
NURSING UNITS AND OTHER PATIENT CARE AREAS SURGERY AND CRITICAL CARE									
Critical care patient care station (2.2–2.6.2)	NR	2	6	NR	No	Yes	8/14	30–60	70–75/21–24
Operating room (2.2–3.3.2) (m), (o)	Positive	4	20	NR	No	Yes	8/14	20–60	68–75/20–24
Procedure room (3.7–3.2) (o), (d)	Positive	3	15	NR	No	Yes	13/NR	20–60	70–75/21–24
Sterile processing room (2.2–3.3.6.13)	NR	2	6	NR	No	Yes	8/14	NR	NR
INPATIENT NURSING									
All anteroom (2.1–2.4.2.3) (u)	(e)	NR	10	Yes	No	Yes	8/14	NR	NR
All room (2.1–2.4.2) (u)	Negative	2	12	Yes	No	Yes	8/14	Max 60	70–75/21–24
Labor/delivery/recovery/postpartum (LDRP) (2.2–2.11.3) (s)	NR	2	6	NR	NR	Yes	8/14	Max 60	70–75/21–24
Patient care area corridor	NR	NR	2	NR	NR	Yes	8/14	NR	NR
Food and supply storage (2.1–4.3.8.11)	NR	NR	2	NR	No	No	8/NR	NR	72–78/22–26
Toilet room (2.1–4.3.9.1)	Negative	NR	10	Yes	No	Yes	8/NR	NR	72–78/22–26

According to ASHRAE 170 Table 7-1

1) SUPPLY AIRFLOW (CFM_F)

$$CFM_F = \frac{ACH \times Volume\ (m^3)}{1.7}$$

Note: The value 1.7 is a constant per ASHRAE 170 Table 7-1.

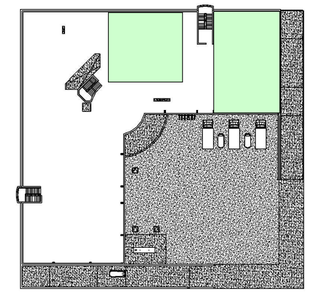
2) EXHAUST AIRFLOW (CFM_{EX})

$$CFM_{EX} = CFM_F + CFM_{in} - CFM_{out}$$

DEFINITIONS

CFM_F	Supply airflow (CFM)
CFM_{EX}	Exhaust airflow (CFM)
ACH	Air Changes per Hour (1/hr)
Volume (m^3)	Room Volume in cubic meters (m^3)
CFM_{in}	Airflow coming into the space (CFM)
CFM_{out}	Airflow going out of the space (CFM)

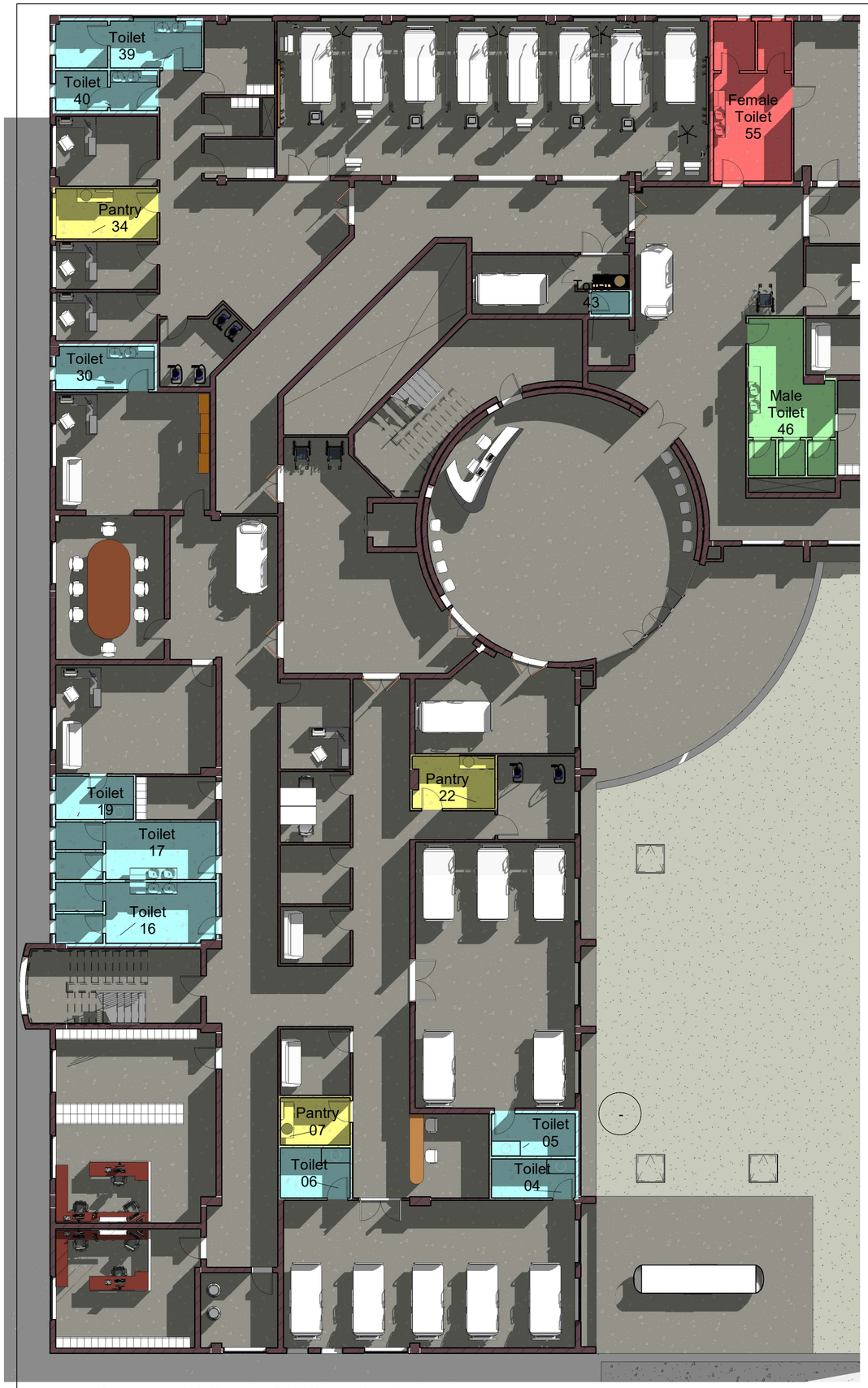
Key Plan:



Legends:

MOHAMED SHABAN
MECHANICAL ENGINEER

Project:	Hospital
Owner:	ENGX
Date:	18/9/2026
Designer:	M.SHABAN
Checked by:	Taha Hussein
DRAWING:	Ground Floor
DISCIPLINE:	HVAC
Scale:	1:50



1 Ground Floor Toilet & Pantry

1 : 150

Table 7.1 Design Parameters—Inpatient Spaces—Hospital Spaces

Function of Space (ad)	Pressure Relationship to Adjacent Areas (n)	Minimum Outdoor ach	Minimum Total ach	All Room Air Exhausted Directly to Outdoors (j)	Air Recirculated by Means of Room Units (a)	Unoccupied Turndown	Minimum Filter Efficiencies (ab)	Design Relative Humidity (k), %	Design Temperature (l), °F/°C
Toilet room (2.1–4.3.9.1)	Negative	NR	10	Yes	No	Yes	8/NR	NR	72–78/22–26
Nourishment area or room (2.1–2.6.7)	NR	NR	2	NR	NR	Yes	8/14	NR	NR

According to ASHRAE 170 Table 7-1

$$CFM_F = (ACH \times Volume(m^3)) / 1.7$$

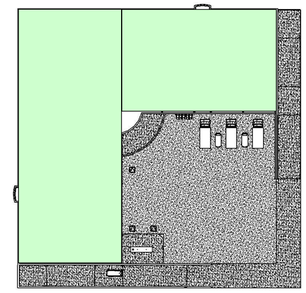
$$CFM_{EX} = 1.1 \times CFM_F$$

Space Name	Space Number	Area(m^2)	Height (m)	ACU	CFM_S	CFM_{EX}	Pressure(pa)
Toilet	04	6	2.75	10	97	106	Negative
Toilet	05	6.5	2.75	10	105	115	Negative
Toilet	06	6.3	2.75	10	100	110	Negative
Toilet	16	17.3	2.75	10	280	308	Negative
Toilet	17	16.4	2.75	10	235	260	Negative
Toilet	19	6	2.75	10	97	106	Negative
Toilet	30	7.6	2.75	10	122	135	Negative
Toilet	39	12.3	2.75	10	200	220	Negative
Toilet	40	7.3	2.75	10	120	132	Negative
Toilet	43	1.48	2.75	10	24	27	(-15)
Male Toilet	46	20.6	2.75	10	330	363	Negative
Female Toilet	55	22.3	2.75	10	360	400	Negative

Space Name	Space Number	Area(m^2)	Height (m)	ACU	CFM_S	CFM_{EX}	Pressure(pa)
Pantry	07	6	2.75	10	97	106	Negative
Pantry	22	7.5	2.75	10	120	130	Negative
Pantry	34	9.2	2.75	10	150	165	Negative

- Female Toilet
- Male Toilet
- Pantry
- Toilet

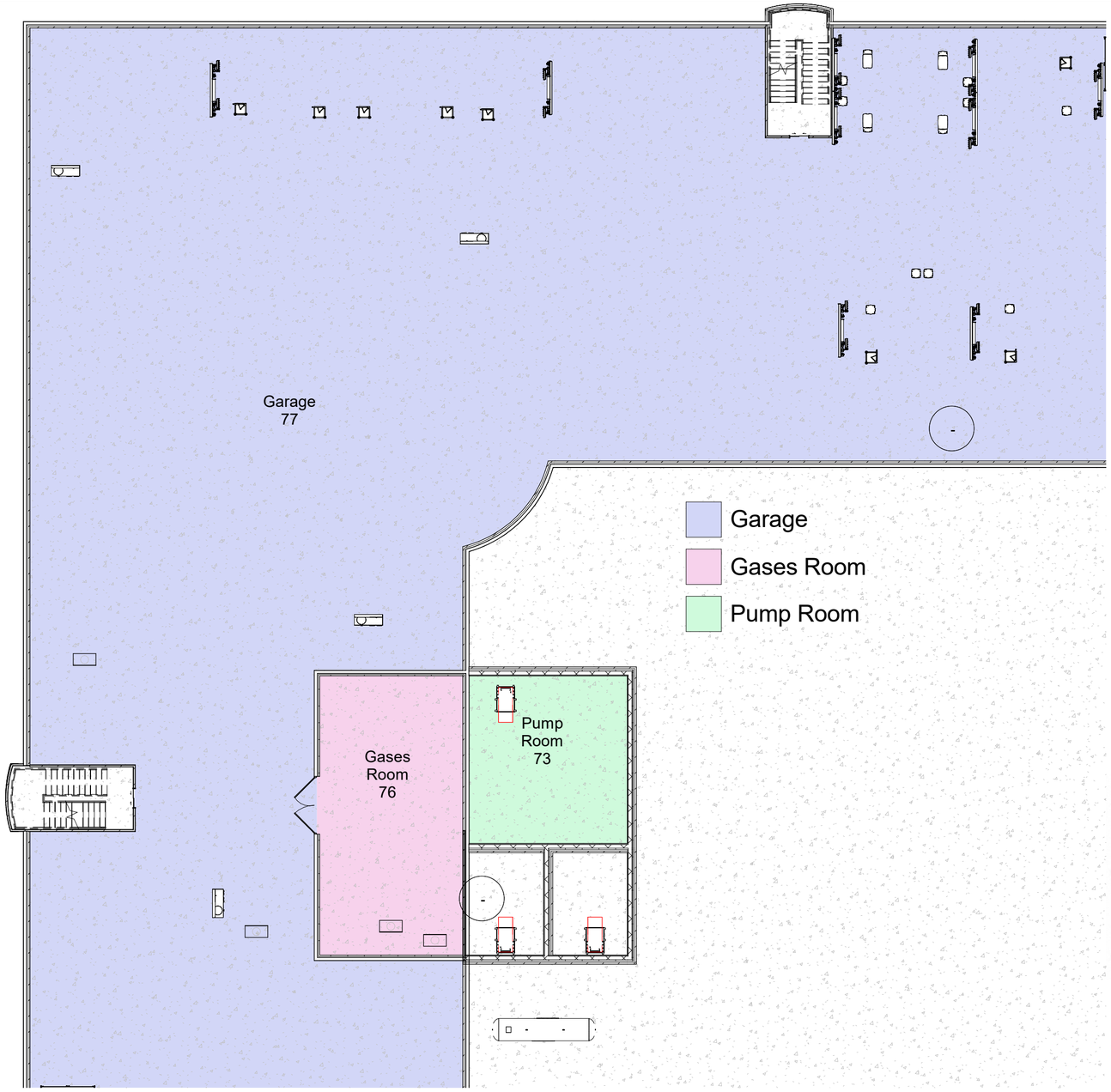
Key Plan:



Legends:

MOHAMED SHABAN
MECHANICAL ENGINEER

Project:	Hospital
Owner:	ENGX
Date:	18/9/2026
Designer:	M.SHABAN
Checked by:	Taha Hussein
DRAWING:	Ground Floor
DISCIPLINE:	HVAC
Scale:	1:50



1 Basement Ventilation

1 : 200

ج- يجب أن تكون معدلات التهوية وسحب الدخان طبقا للنوتة الحسابية التفصيلية والخاصة بكل من التهوية وسحب دخان الحريق على ألا تتجاوز المعدلات الآتية:

- ألا تقل معدلات التهوية بأي حال من الأحوال عن ٧,٥ لتر في الثانية لكل متر مربع من مساحة الجراج الإجمالية شاملة المرافق الداخلية وذلك للجراجات حتى ارتفاع ٤,٥ م، وإذا زاد الارتفاع عن ٤,٥ م فإنه يتم الأخذ بمعدل سريان هواء يكافئ تغيير كامل حجم الهواء بمعدل لا يقل عن ٦ مرات بالساعة لحجم هواء الجراج وفي كلا الحالتين يجب تقديم نوتة حسابية بالحمل التشغيلي الأقصى للسيارات (أيهما أكبر).
- ألا تقل معدلات سحب الدخان بأي حال من الأحوال عن ١٠ لتر في الثانية لكل متر مربع من مساحة الجراج الإجمالية شاملة المرافق الداخلية وذلك للجراجات حتى ارتفاع ٤,٥٠ م، وإذا زاد الارتفاع عن ٤,٥٠ م فإنه يتم الأخذ بمعدل سريان هواء يكافئ تغيير الهواء بمعدل ١٠ مرات بالساعة لحجم الجراج وفي كلا الحالتين يجب تقديم نوتة حسابية بحمل الحريق الأقصى للسيارات (أيهم أكبر).

According to ASHRAE 170 Table 7-1

$$CFM_F = (ACH \times Volume(m^3)) / 1.7$$

$$CFM_{EX} = 1.1 \times CFM_F$$

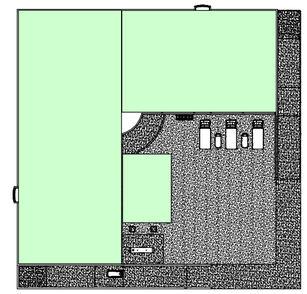
Table 3-3 *Ventilation Design Parameters*

Function of Space	Pressure Relationship to Adjacent Areas (n)	Minimum Outdoor ACH, ach	Minimum Total ACH, ach	All Room Air Exhausted Directly to Outdoors (j)	Air Recirculated by Means of Room Units (a)	RH (k) (%)	Design Temperature (l), (°F/°C)
SURGERY AND CRITICAL CARE							
Medical/Anesthesia gas storage (r)	Negative	N/R	8	Yes	N/R	N/R	N/R

Space Name	Space Number	Area(m^2)	Height (m)	ACU	CFM_S	CFM_{EX}	Pressure(pa)
Pump Room	73	76.4	4.4	10	1977	2174	Negative
Gases Room	76	115	4.4	10	2975	2825	Negative

Space Name	Space Number	Area(m^2)	Height (m)	ACU		$CFM_{General}$	CFM_{Fire}	CFM_{EX}		Pressure(pa)
				General	Fire			General	Fire	
Garage	34	1977	4.4	6	10	30702	51170	33772	56287	Negative

Key Plan:



Legends:

MOHAMED SHABAN
MECHANICAL ENGINEER

Project:	Hospital
Owner:	ENGX
Date:	18/9/2026
Designer:	M.SHABAN
Checked by:	Taha Hussein
DRAWING:	Ground Floor
DISCIPLINE:	HVAC
Scale:	1:50